

Spin Screening in MoS₂ Mirror Twin Boundaries on graphene on Ir(110)

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Mirror twin boundaries (MTBs) are line defects that host confined states in the bandgap of monolayer MoS₂. The confined states closest to the Fermi energy split into a singly and a doubly occupied level separated by the Coulomb repulsion U , forming a spin-1/2 system. The spin of the singly occupied state is then screened by the electron bath present in the substrate below, giving rise to a Kondo resonance pinned to the Fermi energy (1). In this work, we investigate spin screening in MoS₂ MTBs grown on graphene/Ir(110) using low-temperature scanning tunnelling microscopy and spectroscopy. STM imaging reveals well-defined MTBs, while STS spectra recorded along the boundaries show two pronounced impurity peaks separated by the Coulomb energy U , together with a zero-bias feature which can be attributed to the Kondo resonance. Compared to MTBs in MoS₂/Gr/Ir(111) (1), MTBs in MoS₂/Gr/Ir(110) exhibit much higher Kondo intensity and significantly broadened impurity-level widths, indicating strong hybridization γ with the electron bath provided by the graphene/Ir(110) substrate. This enhanced hybridization results in a Kondo temperature (T_K) comparable to the measurement temperatures (0.4 K–7 K), enabling the study of the phase transition between the Kondo singlet phase below and the magnetic phase above the Kondo temperature.

[1] C. van Efferen, J. Fischer, T. A. Costi, A. Rosch, T. Michely, W. Jolie, *Nat. Phys.* **20**, 82 (2024).